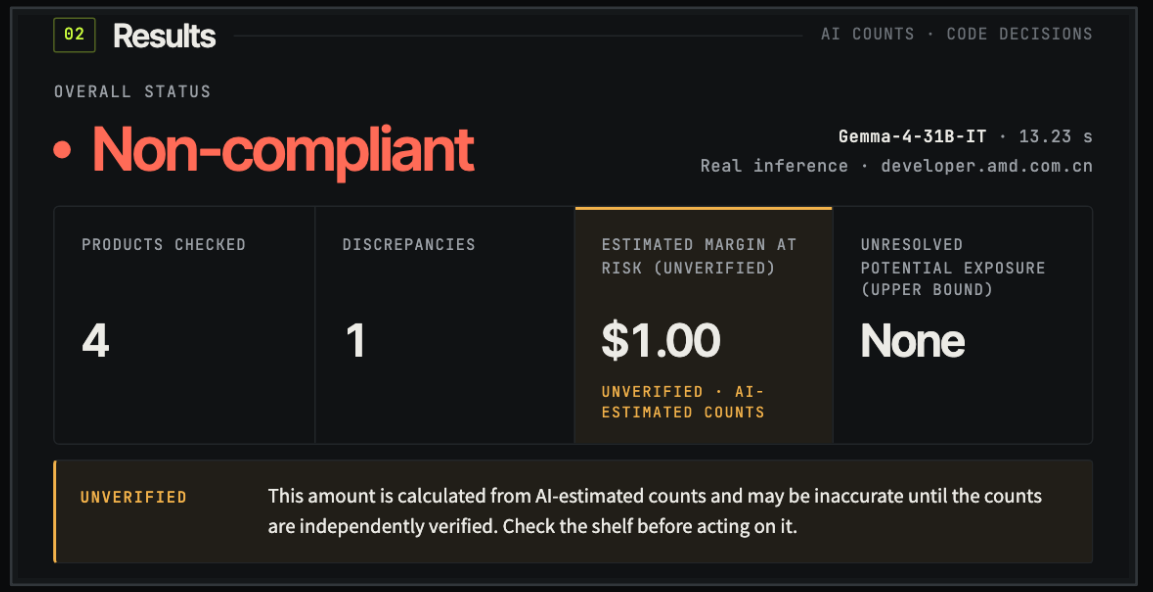
[AMD DEVELOPER HACKATHON: ACT III · REINVENT COMMERCE]

Shelf-Sense

AI-powered retail shelf intelligence

Photograph a shelf. Get per-product counts, a compliance verdict, an estimated margin at risk and a restocking list. The model counts; code decides.



Real result from the app: one inspection on the AMD endpoint

Gemma-4-31B-IT · AMD Developer Cloud Token Factory

[01] THE PROBLEM

Shelves are still checked by eye

Missing facings quietly cost margin. Manual shelf walks are slow and inconsistent, and small stores can’t justify dedicated shelf-scanning hardware.

[01] MANUAL

01

Every count is a person with a clipboard

Checking a bay means walking it, counting facings and writing them down.

[02] INFREQUENT

02

Gaps sit unnoticed between walks

A missing facing can stay missing until the next scheduled check.

[03] INCONSISTENT

03

Two people, two different counts

Half-hidden boxes and variants are judged differently each time.

[04] UNPRICED

04

Nobody knows what the gap costs

Without a number attached, restocking competes with everything else.

[02] THE SOLUTION

One photo in, a preliminary compliance report out

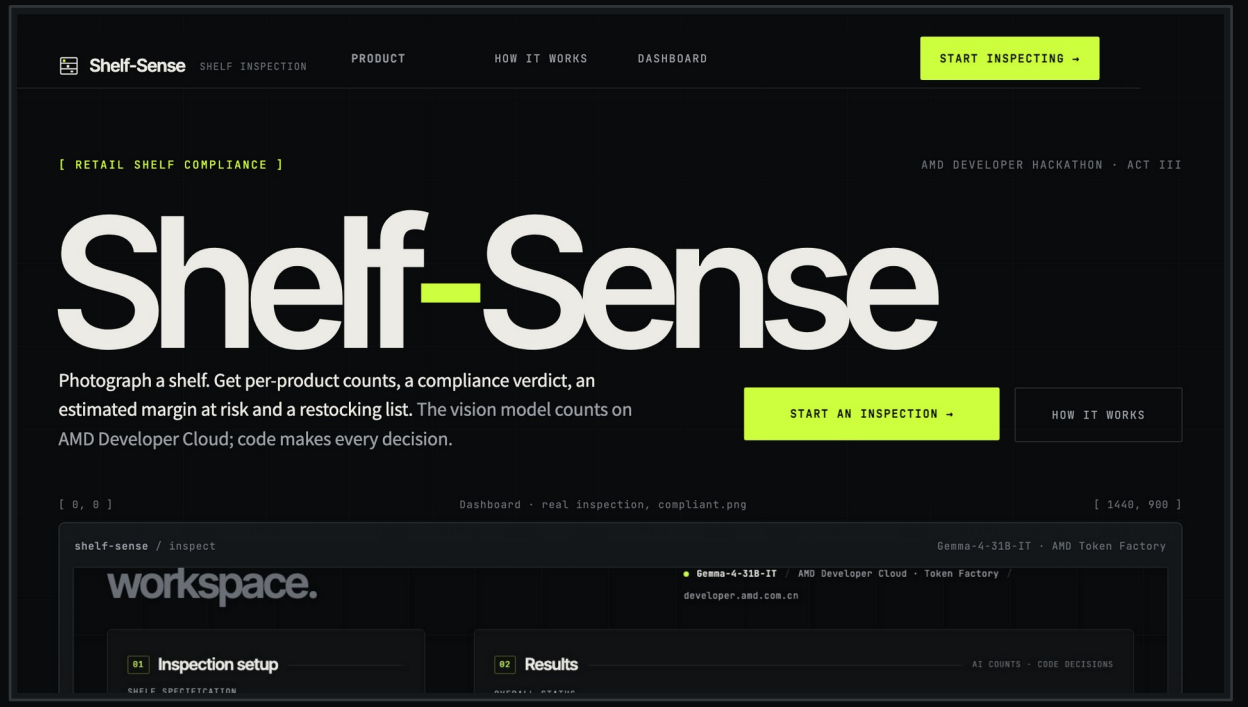
[01] **Choose a shelf specification**

[02] **Add a shelf photograph**

[03] **Run one inspection**

WHAT COMES BACK

- Per-product status and shelf-by-shelf counts
- Discrepancies against the expected inventory
- Estimated Margin at Risk (Unverified), computed in code
- A restocking note, labelled with who wrote it



The live web app: landing page, dashboard and method pages

[03] PRODUCT DEMONSTRATION

A real inspection, end to end

1 Specification | 2 Shelf image | 3 Inference on AMD | 4 Results | 5 Action note

SHELF PHOTOGRAPH — JPEG / PNG · UP TO 15 MB



compliant.png

1354 × 1024 px · 2.4 MB · demo photo

compliant.png, a self-taken photo of a 4-shelf cereal bay

DISCREPANCIES — 1 OF 4 PRODUCTS

Nestle Koko Krunch cereal (any variant) — 7 of 8 · -1 · PARTIAL

PRODUCT RESULTS — COUNTS FROM THE MODEL · STATUS AND MONEY FROM CODE

PRODUCT	EXP.	DET.	Δ	PER SHELF ↓	STATUS	EST. MARGIN*
Nestle Stars cereal (blue box)	16	16	±0	6 / 5 / 3 / 2	PRESENT	$0.00
Nestle Koko Krunch cereal (any variant)	8	7	-1	2 / 1 / 2 / 2	PARTIAL	$1.00
Nestle Milo cereal (green box, any variant)	4	4	±0	1 / 1 / 1 / 1	PRESENT	$0.00
Trix cereal (red box)	4	4	±0	0 / 0 / 2 / 2	PRESENT	$0.00

* Estimated margin at risk, unverified: calculated by code from the model's counts.

Per-shelf counts against expected; Koko Krunch 7 of 8 flagged

Recommended actions — AMD-WORDED · GEMMA-4-31B-IT

Cereal bay, 4 shelves: 1 item needs attention.
Focus on restocking the cereal bay.

01 Nestle Koko Krunch cereal (any variant)
Restock the item.
partially stocked (7 of 8 expected detected); 1 unit short; estimated margin at risk $1.00

Estimated Margin at Risk (Unverified): $1.00.
This figure is calculated from AI-estimated counts that have not been independently verified; check the shelf before relying on it.

NEXT restock 1 item, then re-run the inspection to check the shelf again.

Action wording by Gemma-4-31B-IT via the AMD Token Factory endpoint (2.32s). Product names, counts, statuses and figures are inserted by code, not the model.

AMD-worded note; names, counts and figures inserted by code

[04] TECHNICAL ARCHITECTURE

One Python process, one model call

CODE

Streamlit app

Spec, photo, results UI; three pages

CODE

Inspection + prompt

EXIF-rotate, resize to 1024 px; spec as text

MODEL

AMD Token Factory

Gemma-4-31B-IT returns per-shelf counts (JSON)

CODE

Validation

Pydantic schema, one retry, review flags

CODE

Financial calculation

Verdict and margin estimate, Decimal maths

CODE + MODEL

Action note

Template; AMD may reword; numbers from code

Model output: per-product, per-shelf counts and optional wording. **Code:** validation, statuses, the verdict, every dollar figure, and the checks on any AI wording.

Single Docker container on Render · no database · no other model provider · 195 offline tests

Measured on a real shelf, reported as measured

EXPERIMENT	CHANGE	EXACT	ERROR	SHELF RIGHT
Baseline	First prompt	1 / 4	8 boxes	0 of 2
1 · production	Shelf-by-shelf prompt	3 / 4	1 box	0 of 3
2	1536 px instead of 1024	3 / 4	1 box	0 of 6
3	MiniCPM-V instead of Gemma	3 / 4	1 box	0 of 2
4	Per-shelf crops (by hand)	2–3 / 4	2–3	3 of 3



Shelf 2 of the real photo. Hand count: 2 Koko Krunch. Model: 1, at 100% confidence. Box outlined by hand.

MEASURED

Production misses one half-hidden box in all 9 recorded runs, so the full shelf is reported short.

EXPERIMENTAL

Per-shelf crops fixed the verdict (3 of 3) but overcounted neighbours. Not adopted.

DEMO ASSUMPTIONS

$1.00 per-unit margins. One self-taken photo: a case study, not a benchmark.

FUTURE

Occlusion-aware counting, human verification, many more shelves.

Every inspection runs on AMD

18.2 s

median inspection on the shared endpoint (9 recorded runs, 11.7–45.8 s)

1

vision call per inspection, plus 1 text call to word the actions

24

experiment records kept with the model’s raw replies

MODEL	Gemma-4-31B-IT, vision input
ENDPOINT	AMD Developer Cloud Token Factory, OpenAI-compatible chat completions
EVIDENCE	First raw HTTP 200 and every experiment reply kept in the repository
NOT CLAIMED	We call a hosted service: no GPU, vLLM or ROCm claims

Intended value: a fast, photo-based second opinion for store staff on where to look and roughly what is at stake, without shelf-scanning hardware.

[07] ROADMAP · PLANNED, NOT BUILT

What comes next

[01] **Better product counting under occlusion**

[02] **Human verification of uncertain findings**

[03] **Evaluation on diverse shelf images**

[04] **Reliability testing in a real store**

Shelf-Sense

AMD-hosted vision. Decisions in code.
Honest about its limits.

github.com/bhavya-jpg/shelf-sense

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